DS 3010

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Module 3: Model Selection

Part 1: Introduction

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Motivation for model selection here

Up to this point, we know how to fit a model for a set of p predictors $X_1, \ldots, X_p$, evaluate its performance, and carry out inference procedures.

- How do we know which subset of predictors are important?
- Can we streamline our model?
- Data acquisition can be expensive.
- Bias-variance tradeoff.
- Interpretability.

Model Selection

The process of removing “irrelevant” or “less important” predictors is called:

- Model selection
- Feature selection/ feature screening.
- Variable selection

We know that we cannot just look at a model with all p predictors and remove non-significant predictors (why?). We need some more systematic techniques.

Possible Models

Suppose we are considering 3 potential predictors X_1, X_2 and X_3. How many possible models are there?

- 0 predictor: intercept only
- 1 predictor: X_1, X_2, X_3
- 2 predictors: (X_1, X_2), (X_1, X_3), (X_2, X_3)
- 3 predictors: (X_1, X_2, X_3)
- Total possible models: $8 = 2^3$

p predictors : Total possible models: 2^p

- 10 predictors: $2^{10} = 1024$
- 20 predictors: $2^{20} = 1048576 > 1\text{million}$

Model Selection Strategy

P is small (p < 30)

- Best subset selection

P is large (p >= 30)

- Forward selection
- Backward selection
- Hybrid approach

Best Subset Selection (p <30)

Algorithm:

Step 1

k =0: no predictors , call the model M_0

Step 2

k =1: $X_1, X_2, X_3,$ find best model and call it M_1

k = 2, $(X_1, X_2), (X_1, X_3), (X_2, X_3),$ find best model and call it M_2

k = 3, (X_1, X_2, X_3)……. find best model and call it M_3

…….

k = p, $(X_1, X_2, X_3, \ldots. X_p)$ find best model and call it M_k

Step 3

Select a single best model among M_0, M_1 …. M_p using some criteria.

What criteria should we use ?

Best Subset Selection (p <30)

- This is an exhaustive search. It will produce an optimal model based on some criteria.
- Computationally infeasible for values of p greater than around 40, even with extremely fast modern computers.
- Only work for least squares linear regression

Critiera to Pick Final Model

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i) \text{ ^2}$$

How about model with lowest RSS ?

- RSS always decreases as we add more predictors to the model.
- The full model will all predictors will always be chosen.

For you to think about

Example 1

M_3: $Y \sim X_1 + X_2 + X_3$

M_4: $Y \sim X_1 + X_2 + X_3 + X_4$

Example 2

M_3: $Y \sim X_5 + X_{11} + X_{23}$

M_4: $Y \sim X_1 + X_2 + X_3 + X_4$

Is it true that $RSS_3 > RSS_4$?

This is only true with nested models (when the smaller model contains an exact subset of predictors in the larger model) like Example 1.

Critieria to Pick Final Model

- Mallow's Cp,
- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)
- Adjusted R^2.

Intuition

Criteria

- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)
- Mallow's Cp,
- Adjusted R^2.

- $\text{AIC} = n\log\frac{RSS}{n} + 2p$
- $\text{BIC} = n\log\frac{RSS}{n} + p\log(n)$
- Mallow's $C_p = \frac{RSS}{\sigma^2} - n + 2p$
- Adjusted $R^2 = 1 = \frac{RSS/(n-(p+1))}{TSS/(n-1)}$.

Example

n = 100, **M_3**: $Y \sim X_1 + X_2 + X_3$. **VS** **M_4**: $Y \sim X_1 + X_2 + X_3 + X_4$

p = 4 (intercept included) p = 5 (intercept included)

	RSS	$\boldsymbol{n * log \frac{RSS}{n}}$	**2*p**	**AIC**	**Better model**
M_3	200	$100 * log \frac{200}{100} = 69.31$	2*4 = 8	69.31+8 = 77.31	--
M_4 (A)	198	$100 * log \frac{198}{100} = 68.31$	2*5 = 10	68.31+10 = 78.31	M3
M_4 (B)	195	$100 * log \frac{195}{100} = 66.78$	2*5 = 10	66.78+10 = 76.78	M4

Penalty term

Only keep a predictor in the model if the decrease in RSS is large enough to compensate for the penalty.

Example

	RSS	$n * log \frac{RSS}{n}$	2*p	AIC	Better model
M_3	200	$100 * log \frac{200}{100} = 69.31$	2*4 = 8	69.31+8 = 77.31	--
M_4	198	$100 * log \frac{198}{100} = 68.31$	2*5 = 10	68.31+10 = 78.31	M3
M_4	195	$100 * log \frac{195}{100} = 66.78$	2*5 = 10	66.78+10 = 76.78	M4

	RSS	$n * log \frac{RSS}{n}$	p*log(n)	BIC	Better model
M_3	200	$100 * log \frac{200}{100} = 69.31$	4*log(100) = 18.42	69.31+18.42 = 87.73	--
M_4	198	$100 * log \frac{198}{100} = 68.31$	5*log(100) = 23.03	68.31+23.03 = 91.34	M3
M_4	195	$100 * log \frac{195}{100} = 66.78$	5*log(100) = 23.03	66.78+23.03 = 89.81	M3

Heavier Penalty

Criteria

- **AIC used more frequently**
  - Assumes true model not in candidate pool $M_0, \ldots, M_p$
  - Tries to mimic the true model.

- **BIC has a heavier penalty term.**
  - Will usually pick a simpler/more parsimonious model.
  - BIC Is consistent: if true model is among candidate pool it will eventually lead you to the true model (if n is large enough).

Will these 4 criteria pick the same final model when p is the same?

M_A: $Y \sim X_1 + X_2 + X_3 + X_4 + X_5$

M_B: $Y \sim X_6 + X_7 + X_8 + X_9 + X_{10}$

M_C: $Y \sim X_1 + X_2 + X_7 + X_9 + X_{10}$

p = 6 (intercept included)

	RSS	$n * log \frac{RSS}{n}$	2*p	AIC	Better model
M_A	200	$100 * log \frac{200}{100} = 69.31$	2*6= 12	69.31+12 = 81.31	M_C
M_B	198	$100 * log \frac{198}{100} = 68.31$	2*6 = 12	68.31+12 = 79.31	
M_C	195	$100 * log \frac{195}{100} = 66.78$	2*6 = 12	66.78+12 = 78.78	

Same Penalty

With same p , penalty term is same, the criteria just depends on RSS.

	RSS	$n * log \frac{RSS}{n}$	2*p	AIC	Better model
M_A	200	$100 * log \frac{200}{100} = 69.31$	2*6= 12	69.31+12 = 81.31	M_C
M_B	198	$100 * log \frac{198}{100} = 68.31$	2*6 = 12	68.31+12 = 79.31	
M_C	195	$100 * log \frac{195}{100} = 66.78$	2*6 = 12	66.78+12 = 78.78	

	RSS	$n * log \frac{RSS}{n}$	p*log(n)	BIC	Better model
M_A	200	$100 * log \frac{200}{100} = 69.31$	6*log(100) = 27.63	69.31+27.63 = 96.94	M_C
M_B	198	$100 * log \frac{198}{100} = 68.31$	6*log(100) = 27.63	68.31+27.63 = 95.94	
M_C	195	$100 * log \frac{195}{100} = 66.78$	6*log(100) = 27.63	66.78+27.63 = 94.41	

If the competing models have the same number of predictors p, will all four selection criteria choose the same model?

All criteria lead to same model:

With same p , penalty term is same, the criteria just depends on RSS.

Algorithm

Step 1

k = 0: no predictors , call the model M_0

Step 2

k =1: $X_1, X_2, X_3, \ldots. X_p$ find best model M_1

k = 2, $(X_1, X_2), (X_1, X_3), (X_2, X_3), \ldots$ find best model and M_2

k = 3, (X_1, X_2, X_3)……. find best model and M_3

……

k = p, $(X_1, X_2, X_3, \ldots. X_p)$ find best model M_k

RSS

Step 3

Select a single best model among $M_0, M_1 \ldots. M_p$ using criteria: **AIC, BIC…..**

Step 2: best model is the model with lowest RSS.

- Fit the model with **the entire dataset**
- Calculate **RSS**, RSS = $\sum_1^n (yi - \hat{y}_i)^2$
- Best model **M_i : model** with lowest **RSS.**

Algorithm:

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k = 3, (X_1, X_2, X_3)……. find best model and M_3

……

k = p, $(X_1, X_2, X_3, \ldots. X_p)$ find best model M_k

Step 3

Select a single best model among M_0, M_1 **….** M_p using some criteria.

Step 2: best model is the model with lowest RSS.

Example: when k = 2, best model M_2 is the one with lowest RSS

Model	(X_1, X_2)	(X_1, X_3)	(X_1, X_4)	(X_1, X_5)	…. .	(X_{p-1}, X_p)
RSS	746	557	349	638	953	596

Step 3: Final best model is the model with lowest AIC/BIC/Mallow's CP or highest Adjusted R square.

Lowest

Model	M_0	M_1	M_2	M_3	$M_{...}$	M_k
AIC	76	57	39	38	53	96
BIC	82	68	42	57	65	104
CP	…	…	…	…	…	…
Adj R^2	0.95	0.97	0.86	0.99	0.84	0.78

- AIC, BIC, Mallow's CP: the smaller, the better
- Adjusted R square: the larger, the better

If the competing models have **different** number of predictors p, will all four selection criteria choose the same model?

No

In order to select the best model with respect to test error, we need to estimate this test error.

1. We can indirectly estimate test error by making an adjustment to the training error to account for the bias due to overfitting. E.g. AIC, BIC… (Note, so far, in best subset selection, each model is fit using the entire dataset)
2. We can directly estimate the test error, using either a validation set approach or a cross-validation approach.
   - **Validation set approach**
   - **Cross-Validation**

Best Subset Selection + Validation Set Approach

Step 0: Split dataset into training (50%) and test (50%) dataset.

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k = 2, $(X_1, X_2), (X_1, X_3), (X_2, X_3), \ldots$ find best model and M_2

k = 3, (X_1, X_2, X_3)……. find best model and M_3

……

k = p, $(X_1, X_2, X_3, \ldots. X_p)$ find best model M_k

Training MSE

Step 3

Select a single best model among $M_0, M_1 \ldots. M_p$ using **test MSE.**

Step 2: best model is the model with lowest training RSS.

- Fit the model with **training data**
- Calculate **training MSE.**
- Best model **M_i : model** with lowest **training MSE.**

Step 3: Final best model is the model with lowest test MSE.

Best Subset Selection + Validation Set Approach

Drawback of Validation Set Approach:

- Don’t get to see all the data.
- Randomness

In-class activity:

Best Subset Selection + K-fold Cross Validation

Best Subset Selection + K-fold Cross Validation

Divide the dataset into k folds

Outer for i:

- Use fold **i** for test, and the rest folds for training
- Perform Best Subset Selection on Training Data and obtained 19 best subset models
- **Inner for** all **19** best subset models: calculate test_MSE[i, j] (Use matrix)

Model size	1	2	..	19
Test MSE (test: fold i)	45	56	49	89

Model size	1	2	..	19
Test MSE (test: fold 1)	45	56	49	89
Test MSE (test: fold 2)	45	56	49	89
……	…	…	…	…
Average Test MSE	45	56	49	89

Best model size is the model with lowest Average Test MSE